Magma & LMFDB Mini-Course

Hunt 3: Elliptic Curves

Eran Assaf

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We will find a non-negative integer n. Each of the four core problems below has a single non-negative integer answer n_i. The bonus is harder and does not count toward the sum. Set

$$n \ = \ n_2 - n_1 - n_3 - n_4.$$

Solutions in Magma or Sage are both welcome.

Problem 3.1 (Reduction mod p). For the running curve $E = \texttt{37.a1}$ (Weierstrass form $y^2+y = x^3 - x$, conductor 37, rank 1), let $a_p := p+1-\#E(\mathbb{F}_p)$ denote the trace of Frobenius at the prime p of good reduction. Let n_1 be the smallest prime p at which E has good reduction and $a_p = 0$. (Such p are called *supersingular* for E.)

Problem 3.2 (Mordell–Weil rank). The lowest conductors of elliptic curves over $\mathbb{Q}$ of rank $0, 1, 2, 3$ are $11, 37, n_2, 5077$ respectively (Cremona 1992; conditional only on completeness of his tables, now established for $N \leq 500000$). Let n_2 be the smallest conductor of an elliptic curve over $\mathbb{Q}$ of Mordell–Weil rank ≥ 2.

Problem 3.3 (Mazur's theorem). Mazur's theorem says $E(\mathbb{Q})_{\text{tors}}$ is one of exactly 15 isomorphism classes. The largest of these is $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/8\mathbb{Z}$ (16 rational torsion points). Let E be the curve $E/\mathbb{Q}$ with $E(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/8\mathbb{Z}$ having smallest conductor. Let $\overline{\rho}_{E,2} : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}(E[2]) \simeq \mathrm{GL}_2(\mathbb{F}_2)$ be the associated mod-2 Galois representation, and let

$$n_3 = [\mathrm{GL}_2(\mathbb{F}_2) : \mathrm{Im}(\overline{\rho}_{E,2})]$$

be the index of its image.

Problem 3.4 (CM and the Hilbert class polynomial). The *Hilbert class polynomial* $H_D(x) \in \mathbb{Z}[x]$ of an imaginary quadratic discriminant D has degree $h(D)$ and its roots are the CM j-invariants of elliptic curves $E/\mathbb{C}$ with $\mathrm{End}(E) = \mathcal{O}_D$. For $D = -23$, $h(-23) = 3$, so $H_{-23}(x) \in \mathbb{Z}[x]$ is a cubic.

Let α be any root of $H_{-23}(x)$ in $\overline{\mathbb{Q}}$, a fixed algebraic closure of $\mathbb{Q}$. Let n_4 be the absolute discriminant of the number field $\mathbb{Q}(\alpha)$:

$$n_4 \ = \ |\operatorname{disc} \mathbb{Q}(\alpha)|.$$

Bonus 3 (Murmurations, after He–Lee–Oliver–Pozdnyakov 2022). The He–Lee–Oliver–Pozdnyakov phenomenon: order elliptic curves $E/\mathbb{Q}$ by conductor, group by root number $w(E) = \pm 1$ (= parity of analytic rank), and average $a_p(E)/\sqrt{p}$ over each group. The resulting function of p is *not* flat or random – it shows a striking oscillation, the *murmuration*.

(a) Pick a conductor range, e.g. $N \in [7500, 10000]$, and pull from the LMFDB all elliptic curves $E/\mathbb{Q}$ in that range, along with their a_p values for primes $p \in [10, 1000]$ and their root numbers.

(b) For each prime p in your range, separately compute

$$\mu_+(p) \;=\; \frac{1}{|\{E : w(E) = +1\}|} \sum_{E:w(E)=+1} \frac{a_p(E)}{\sqrt{p}}, \qquad \mu_-(p) \;=\; \text{(similarly for } w(E) = -1).$$

(c) Plot $\mu_+(p)$ and $\mu_-(p)$ as functions of p. The murmuration should be clearly visible: the two curves oscillate, with opposite phase. Compare your plot to Figure 1 of HLOP (arXiv:2204.10140).

(d) (Optional, harder.) Compute the murmuration density predicted by the Sawin–Sutherland conjecture (arXiv:2504.12295) for the elliptic-curve case, and overlay it on your plot.